CIE A Level Chemistry

Your notes

8.3 Homogeneous & Heterogeneous Catalysts

Contents

* Types of Catalyst

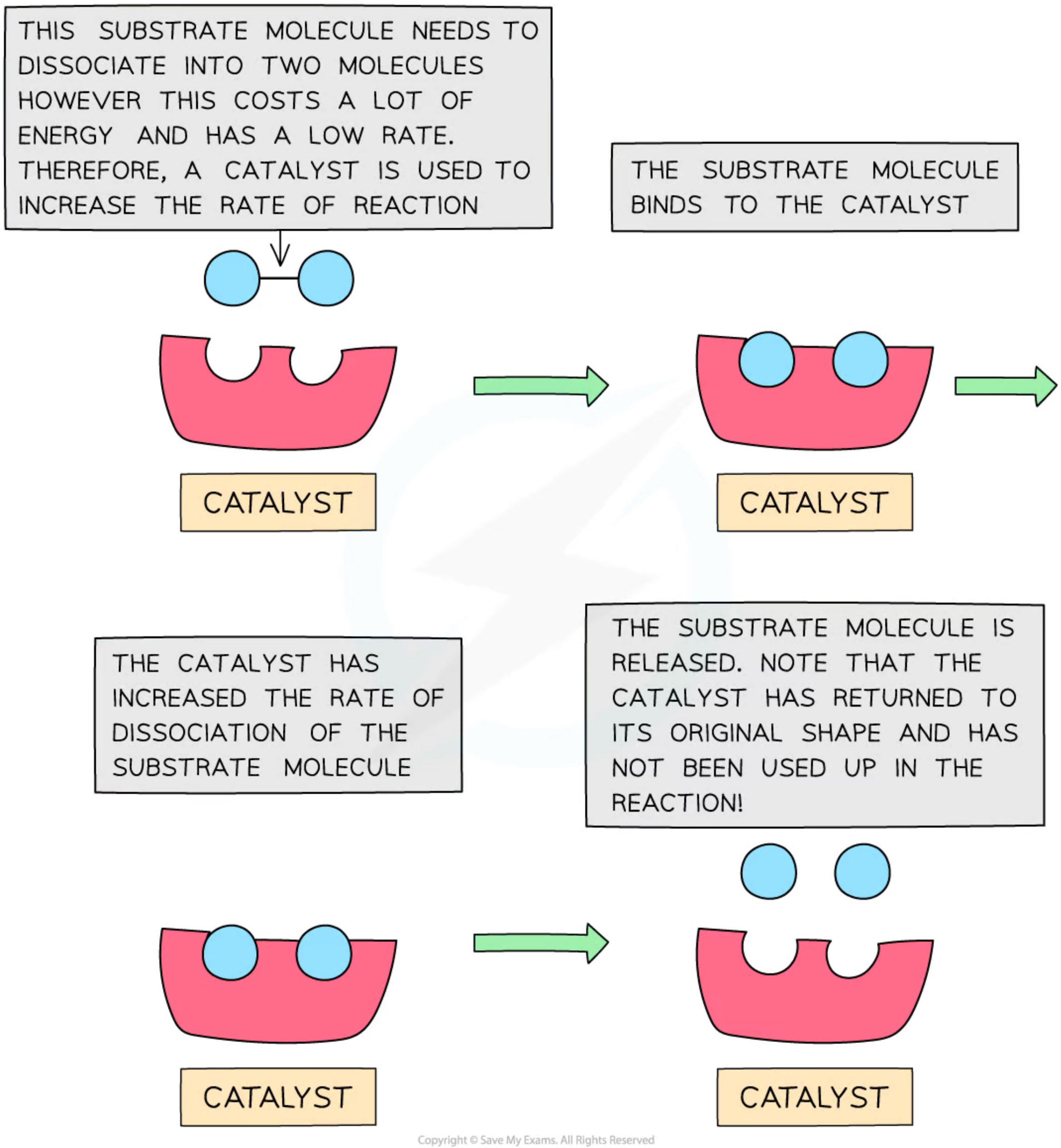
Types of Catalyst

Your notes

Explaining How Catalysts Work

- **Catalysis** is the process in which the rate of a chemical reaction is increased, by adding a substance called a **catalyst**
- A catalyst increases the rate of a reaction by providing the reactants with an **alternative reaction pathway** which is **lower in activation energy** than the uncatalysed reaction
- Catalysts can be divided into two types:
  - Homogeneous catalysts
  - Heterogeneous catalysts
- **Homogeneous** means that the catalyst is in the **same phase** as the reactants
  - For example, the reactants and the catalyst are all liquids
- **Heterogeneous** means that the catalyst is in a **different phase** to the reactants
  - For example, the reactants are gases but the catalyst used is a solid

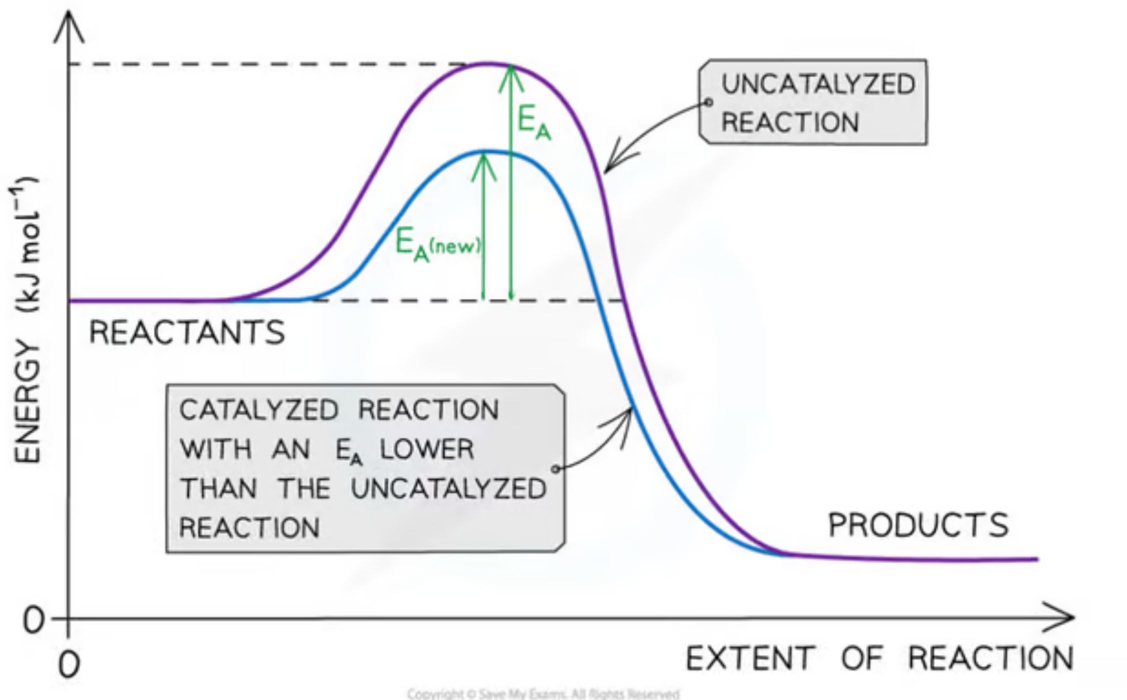
How a catalyst works



The catalyst speeds up a reaction that would normally be slow due to the high activation energy. The catalyst is not used up in the chemical reaction and is not taking part in the chemical reaction

How catalysts affect the reaction pathway

Your notes



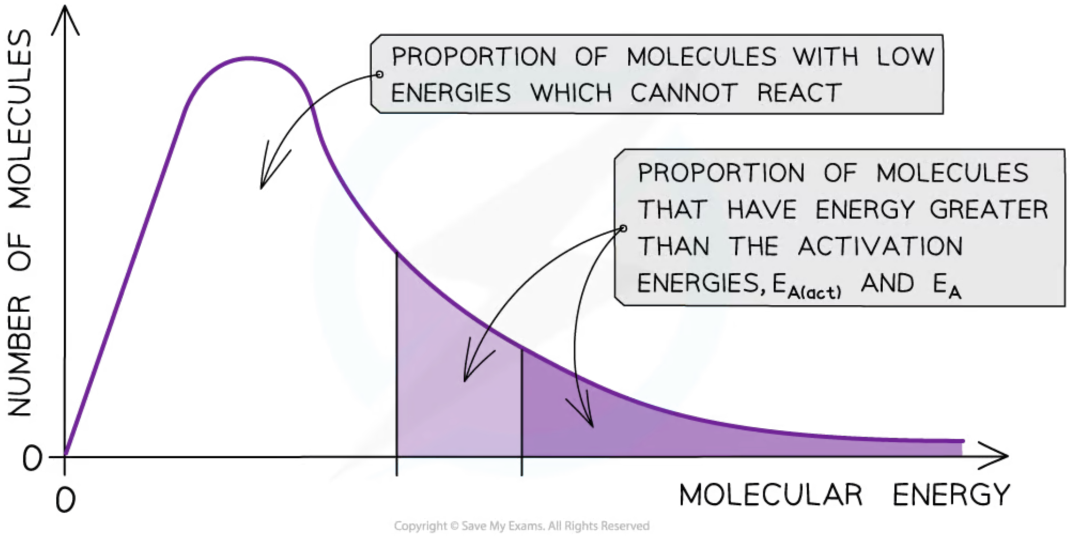
The catalyst allows the reaction to take place through a different mechanism, which has a lower activation energy than the original reaction

Boltzmann distribution curve

- **Catalysts** provide the reactants with another pathway which has a lower activation energy
- By lowering E_a, a **greater proportion** of molecules in the reaction mixture have sufficient energy for an **effective collision**
- As a result of this, the rate of the catalysed reaction is increased compared to the uncatalyzed reaction

How catalysts affect the number of particles with sufficient energy to react (E_a)

Your notes



The total shaded area (both dark and light shading) under the curve shows the number of particles with energy greater than the E_a with a catalyst. This area is much larger than the dark shaded area which shows the number of particles with energy greater than the E_a without a catalyst